

Predicting Disaster Vulnerability Using Radial Basis Function (RBF) Networks with Historical Disaster Data

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Assignment No. 4

SDG Mapping

- **SDG 11 – Sustainable Cities and Communities:** Helps identify vulnerable areas and improve disaster preparedness.
- **SDG 13 – Climate Action:** Supports climate-related risk assessment and disaster mitigation.

1. Problem Understanding

Natural disasters such as floods, cyclones, earthquakes, and landslides can cause significant damage to communities. Historical disaster data can be analyzed to identify patterns and predict the vulnerability of a region.

The objective is to develop an **RBF Network** that predicts the disaster vulnerability level of an area using factors such as rainfall, temperature, population density, elevation, distance from rivers, previous disaster frequency, and infrastructure quality.

Possible output classes:

- Low Vulnerability
 - Medium Vulnerability
 - High Vulnerability
-

2. Dataset

A suitable dataset can contain historical records with the following attributes:

Feature	Description
Rainfall	Average rainfall in mm

Feature	Description
Temperature	Average temperature in °C
Population_Density	People per square kilometer
Elevation	Elevation above sea level
River_Distance	Distance from nearest river
Previous_Disasters	Number of previous disasters
Infrastructure_Index	Quality of infrastructure
Vulnerability	Target vulnerability level

Example:

Rainfall	Temperature	Population Density	Elevation	River Distance	Previous Disasters	Infrastructure	Vulnerability
180	29	5200	15	2.1	5	40	High
90	27	2100	80	12.5	2	70	Medium
45	25	900	250	35.0	0	90	Low
220	31	6500	8	1.2	7	30	High
110	28	3000	60	10.0	3	65	Medium

3. Data Preprocessing

Use **Pandas** for data cleaning and **NumPy** for numerical processing.

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv("disaster_data.csv")

print(df.head())

print(df.isnull().sum())

# Handle missing numerical values

num_cols = [

    "Rainfall", "Temperature", "Population_Density",

    "Elevation", "River_Distance",

    "Previous_Disasters", "Infrastructure_Index"

]

for col in num_cols:

    df[col] = df[col].fillna(df[col].median())

# Remove duplicate records

df = df.drop_duplicates()

print(df.describe())
```

4. Feature and Target Selection

```
X = df[num_cols]

y = df["Vulnerability"]
```

The input variables are the disaster-related factors, while **Vulnerability** is the target variable.

5. Data Normalization

RBF networks use distance calculations, so features should be placed on comparable scales.

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

Standardization converts features approximately to:

where:

- x = original value
- μ = mean
- σ = standard deviation

6. Train-Test Split

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X_scaled,
```

```
    y,
```

```
    test_size=0.2,
```

```
    random_state=42,
```

```
    stratify=y
```

```
)
```

Here, **80%** of the data is used for training and **20%** for testing.

7. RBF Network

An RBF network uses radial basis functions to measure the similarity between an input and selected centers.

The Gaussian RBF function is:

where:

- x = input vector
- μ_i = RBF center
- σ_i = spread parameter

A simple implementation can use K-Means to determine the RBF centers.

```
from sklearn.cluster import KMeans
```

```
from sklearn.metrics import accuracy_score
```

```
from sklearn.linear_model import LogisticRegression
```

```
# Select RBF centers
```

```
kmeans = KMeans(n_clusters=10, random_state=42, n_init=10)
```

```
kmeans.fit(X_train)
```

```
centers = kmeans.cluster_centers_
```

Generate RBF features:

```
def rbf_features(X, centers, sigma=1.0):
```

```
    features = []
```

```
    for center in centers:
```

```
distance = np.linalg.norm(X - center, axis=1)

phi = np.exp(-(distance ** 2) / (2 * sigma ** 2))

features.append(phi)
```

```
return np.array(features).T
```

Transform the data:

```
X_train_rbf = rbf_features(
    X_train, centers, sigma=1.0
)
```

```
X_test_rbf = rbf_features(
    X_test, centers, sigma=1.0
)
```

Train the output layer:

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train_rbf, y_train)
```

```
y_pred = model.predict(X_test_rbf)
```

8. Model Evaluation

```
from sklearn.metrics import (
    accuracy_score,
    classification_report,
```

```
confusion_matrix
)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

print(classification_report(
    y_test,
    y_pred
))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

Important evaluation measures include:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

For disaster prediction, **recall for the High Vulnerability class** is particularly important because failing to identify a highly vulnerable area could have serious consequences.

9. Visualization

A vulnerability distribution can be visualized using Pandas:

```
import matplotlib.pyplot as plt
```

```
df["Vulnerability"].value_counts().plot(  
    kind="bar"  
)
```

```
plt.title("Disaster Vulnerability Distribution")
```

```
plt.xlabel("Vulnerability Level")
```

```
plt.ylabel("Number of Areas")
```

```
plt.show()
```

A confusion matrix can also be visualized:

```
import seaborn as sns
```

```
sns.heatmap(  
    confusion_matrix(y_test, y_pred),  
    annot=True,  
    fmt="d",  
    cmap="Blues"  
)
```

```
plt.title("RBF Network Confusion Matrix")
```

```
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
```

```
plt.show()
```

10. Analysis and Interpretation

The RBF network learns nonlinear relationships between disaster-related factors and vulnerability.

For example:

- **High rainfall + low elevation + short river distance** may indicate higher flood vulnerability.
- **High population density** can increase the potential human impact of disasters.
- **Frequent historical disasters** can indicate recurring risk.
- **Poor infrastructure** can increase vulnerability.
- **Higher elevation and stronger infrastructure** may reduce vulnerability in some disaster scenarios.

The RBF network is useful because disaster vulnerability is often **nonlinear**. A small change in rainfall may have little effect in one region but a major effect in another depending on elevation, infrastructure, and population.

11. SDG Relevance

SDG 11 – Sustainable Cities and Communities

The model can help authorities:

- Identify vulnerable communities.
- Prioritize disaster-resistant infrastructure.
- Improve emergency planning.
- Support safer urban development.
- Allocate emergency resources effectively.

SDG 13 – Climate Action

The model can:

- Analyze historical climate-related disasters.
 - Identify areas experiencing increasing vulnerability.
 - Support climate adaptation strategies.
 - Improve disaster preparedness.
 - Assist governments in developing risk-reduction policies.
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12. Innovation

A useful extension is to develop a **Disaster Vulnerability Dashboard**.

The dashboard could allow a user to enter:

Rainfall

Temperature

Population Density

Elevation

Distance from River

Previous Disasters

Infrastructure Index

The RBF model would then display:

Predicted Vulnerability: HIGH

Risk Score: 87%

Recommended Action:

- Evacuation preparedness
- Emergency resource allocation

- Flood monitoring
- Infrastructure inspection

A geographical map could also display different areas according to their predicted vulnerability.

13. Expected Outcome

The expected outcome is an RBF-based machine-learning system capable of classifying areas into **Low, Medium, or High disaster vulnerability**.

The system demonstrates how historical disaster data, machine learning, and visualization can support **disaster preparedness, sustainable communities, and climate adaptation**.

Conclusion

An **RBF Network** provides an effective approach for predicting disaster vulnerability because it can model complex nonlinear relationships among environmental, demographic, and infrastructure factors. Combining **NumPy, Pandas, RBF modeling, visualization, and historical disaster data** creates a practical analytical solution aligned with **SDG 11 and SDG 13**.